

BRAIDing receptors for cell specific targeting

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Reviewed Preprint

Published from the original preprint after peer review and assessment by eLife.

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Reviewed preprint posted

September 14, 2023 (this version)

Posted to bioRxiv

July 18, 2023

Sent for peer review

June 19, 2023

Abstract

Systemic toxicity is a major challenge to therapeutic development, consequently, cell-type specific targeting is highly desirable to improve on-target cell efficacy while reducing off-target toxicity. Here, we describe a cell targeting system we called BRAID (**BR**idged **A**ctivation by **I**ntra/intermolecular **D**ivision) where an active molecule is divided into two inactive or less active parts that are subsequently brought together via a bridging receptor on target cell. We tested this hypothesis using WNT/ β -catenin signaling system and demonstrated that a multivalent WNT agonist molecule divided into two inactive components induced signaling specifically on hepatocytes assembled via two different epitopes on a hepatocyte receptor, β Klotho. These data provided proof-of-concept for a cell specific targeting approach and also demonstrated the feasibility of combining different signaling pathways where desirable. This approach has broad applications to other receptor systems.

eLife assessment

This **important** study introduces a new therapeutic cell targeting system, where an active molecule is divided into inactive or less active parts that are subsequently linked via a bridging receptor on a target cell. **Solid** experimental proof of concept involving the activation of Wnt signaling in hepatocytes is presented, which could be further strengthened by demonstrating the specificity of the effect when applied to a mix of different cell types. This study may be of substantial interest to medical biologists focused on targeted therapeutics or seeking to utilize cell type-specific pathway modulation in their research.

Introduction

Receptor mediated signaling is fundamental to cell-cell communication, regulation of cellular growth/differentiation, maintenance of tissue function/homeostasis, and regulation of injury repair. When a cell specific response is necessary, nature achieves that selectivity through various mechanisms such as cell specific receptor, ligand gradient, short range ligand, and direct cell-cell contacts. However, most ligand receptor systems have pleiotropic effects due to broad expressions of receptors on multiple cell types. While this may be important when a coordinated response in multiple cell types or tissues is needed, achieving cell specific signaling has been a major challenge

for both research and therapeutic development to avoid systemic toxicity or off-target effects. In addition, it is often difficult to identify differences between normal and diseased cells for a particular signaling pathway of interest, making it challenging, if not impossible, to selectively modulate that signaling pathway either in diseased or normal cells specifically. Therefore, effective cell targeting is needed for both research and therapeutic development.

Comparing to cell targeted antagonists, cell targeted agonists are even more complex to design. In addition to cell targeting, agonists also have stringent requirements on affinity, epitope, and geometry of the molecules (Dickopf et al., 2020 [DOI](#)). Previous efforts to engineer cell selective growth factors and cytokines have employed various approaches such as the “chimeric activators” concept (Cironi et al., 2008 [DOI](#)), where a cell targeting arm (“targeting element”) is attached to either the wild-type ligand or a mutated ligand with reduced affinity toward signaling receptor (“activity element”). Such an approach has been applied to several signaling pathways, such as erythropoietin (Taylor et al., 2010 [DOI](#), Burrill et al., 2016 [DOI](#)), interferons (Cironi et al., 2008 [DOI](#), Garcin et al., 2014 [DOI](#)), and interleukin-2 (Ghasemi et al., 2016 [DOI](#), Lazear et al., 2017 [DOI](#)). While little selectivity could be achieved with attaching a targeting arm to a wild-type ligand, selectivity of up to 1000-fold has been reported when a mutated ligand was used (Garcin et al., 2014 [DOI](#)). The “chimeric activators” approach is based on the cooperativity concept, where mutations that weaken the affinity of a natural ligand to its receptor are selected as the “activity element”. Due to the weakened affinity, the mutant “activity element” alone displays a significantly right shifted dose response curve (or lower potency) in an activity assay, on both target and non-target cells that express the signaling receptors. When the “targeting element” is tethered to the mutant “activity element”, the “targeting element” helps to increase the local concentration of the mutant “activity element” on the desired target cell surface, driving engagement of signaling receptor and subsequent intracellular signaling activation, left shifting the activity dose response curve (increase potency) on the target cell and creating a separation between target vs non-target cells. While conceptually elegant, identifying the appropriate mutations that achieve the precise reduction of affinity that could be rescued by the “targeting element” is not trivial. In some cases, while the potency could be rescued, the maximal signaling strength remains compromised, resulting in a partial agonist. In addition, the selectivity is not exquisite, with most literature examples reporting a modest 10-fold separation between targeting vs non-targeting cells. Additional strategies have also been reported that show promising results, for example, inactive pro-drug requiring tumor-associated proteases to activate (Puskas et al., 2011 [DOI](#), Skrombolas et al., 2019 [DOI](#)), complementation of two inactive components via targeted assembly in trans (Banaszek et al., 2019 [DOI](#)), and passive enrichment on the target cell leading to colocalization of monomeric subunits (Mock et al., 2020 [DOI](#)) or inactive “split” components (Quijano-Rubio et al., 2022 [DOI](#)).

Here, we present a novel concept to achieve cell targeting for ligand/receptor systems that involves multicomponent receptor complexes, where a productive signaling competent receptor complex formation is directly assembled through a “bridging element”. We have tested this concept using the WNT/ β -catenin signaling pathway as a model system.

The WNT pathway is highly conserved across species and crucial for embryonic development, and adult tissue homeostasis and regeneration (Nusse and Clevers, 2017 [DOI](#)). WNT induced signaling through β -catenin stabilization has been widely studied and is achieved by ligand binding to frizzled (FZD) and low-density lipoprotein receptor-related protein (LRP) family of receptors. There are nineteen mammalian WNTs, 10 FZDs (FZD₁₋₁₀), and 2 LRPs (LRP5 and LRP6). WNTs are highly hydrophobic due to lipidation that is required for function and are promiscuous, capable of binding and activating multiple FZD and LRP pairs (Janda et al., 2012 [DOI](#), Kadowaki et al., 1996 [DOI](#), Dijksterhuis et al., 2015 [DOI](#)). Elucidating the functions of individual FZDs in tissues has been hampered by difficulties in producing the ligands and lack of receptor and tissue selectivity. Recent breakthroughs in the development of WNT-mimetic molecules have largely resolved the production and receptor specificity challenges (Janda et al., 2017 [DOI](#), Chen et al., 2020 [DOI](#), Tao et al., 2019 [DOI](#), Miao et al., 2020 [DOI](#)). While tissue selectivity could be partly achieved by tissue injury as

damaged tissues seem more sensitive to WNTs (Xie et al., 2022 [↗](#)), it still represents a significant technical and therapeutic development advancement to be able to target WNTs to specific cells and tissues.

The WNT mimetics reported so far are all bispecific and can simultaneously bind to FZDs and LRPs, and the optimal stoichiometry (at least for the antibody-based molecules) are tetravalent bispecific (2:2 format), requiring two FZD binders and two LRP binders in the same molecule to achieve efficient signaling (Tao et al., 2019 [↗](#), Chen et al., 2020 [↗](#)). We took advantage of the fact that two FZD binders alone and two LRP binders alone do not signal. Cell specificity could be achieved by attaching a “targeting element” (capable of binding to another cell surface receptor, called a bridging receptor here) to these two inactive molecules, and signaling competent receptor complexes consisting of two FZDs and two LRPs could then be assembled via the bridging receptor on the target cell surface. This approach reduces or eliminates the need to mutate and reduce the affinity of the “active elements” toward signaling receptors, and creates a highly cell specific activation of the signaling pathway as the individual components are inactive. The detailed proof of concept studies based on the WNT/ β -catenin signaling pathway are presented herein. We have called the cell targeted WNT mimetics SWIFT (Splitting of WNT to Induce Functional Targeting).

Results

Conceptual design of cell targeted activators via a bridging receptor

Fig. 1A [↗](#) shows one optimized design for a WNT mimetic that is a tetravalent bispecific antibody-based molecule. It is important to highlight that efficient WNT/ β -catenin signaling requires two FZD binding domains and two LRP binding domains in one molecule (Chen et al., 2020 [↗](#)). **Fig. 1B** [↗](#) shows a cell targeting approach using the bridging receptor concept, the SWIFT approach. The first step of this concept is to split the active molecule into inactive components. In the case of the tetravalent bispecific WNT mimetic, the active molecule is split into one molecule having two FZD binding arms and a second molecule having two LRP binding arms. These two inactive components are then each tethered to a different “bridging element” binding to a different epitope on the same bridging receptor (BR, the tethered molecules named α LRP- α BR-1 and α FZD- α BR-2 in **Fig. 1B** [↗](#), top two panels). α LRP- α BR-1 and α FZD- α BR-2 are each inactive on non-targeting cells where the bridging receptor (BR) is not expressed (**Fig. 1C** [↗](#) top panels). On targeting cells where the BR is expressed, FZD and LRP are then assembled or brought into proximity with one another by α LRP- α BR-1 and α FZD- α BR-2 via BR recreating the signaling competent receptor complexes (**Fig. 1B** [↗](#) bottom panel).

Proof of concept with the targeted WNT mimetic molecules

To test the concept shown in **Fig. 1B** [↗](#), we selected β Klotho and endocrine fibroblast growth factor 21 (FGF21) ligand system as the bridging receptor system. FGF21 is an endocrine hormone produced by the liver that regulates metabolic homeostasis (BonDurant and Potthoff, 2018 [↗](#)). FGF21 signals through FGFR1c, FGFR2c, and FGFR3c in the presence of co-receptor β Klotho (Zhang and Li, 2015 [↗](#)). The binding of FGF21 to the receptor complex is primarily driven by its affinity toward β Klotho via its C-terminal domain (Lee et al., 2018 [↗](#), Shi et al., 2018 [↗](#)), while its N-terminal domain is important for FGFR interaction and signaling (Yie et al., 2012 [↗](#), Micanovic et al., 2009 [↗](#)). β Klotho binding antibodies have also been identified that could induce β Klotho/FGFR signaling and one particular agonistic β Klotho antibody binds to a different epitope on β Klotho from FGF21 and does not compete with FGF21 binding (Min et al., 2018 [↗](#)). Therefore, the following bridging receptor (β Klotho) binding elements were selected to test the cell targeting SWIFT concept:

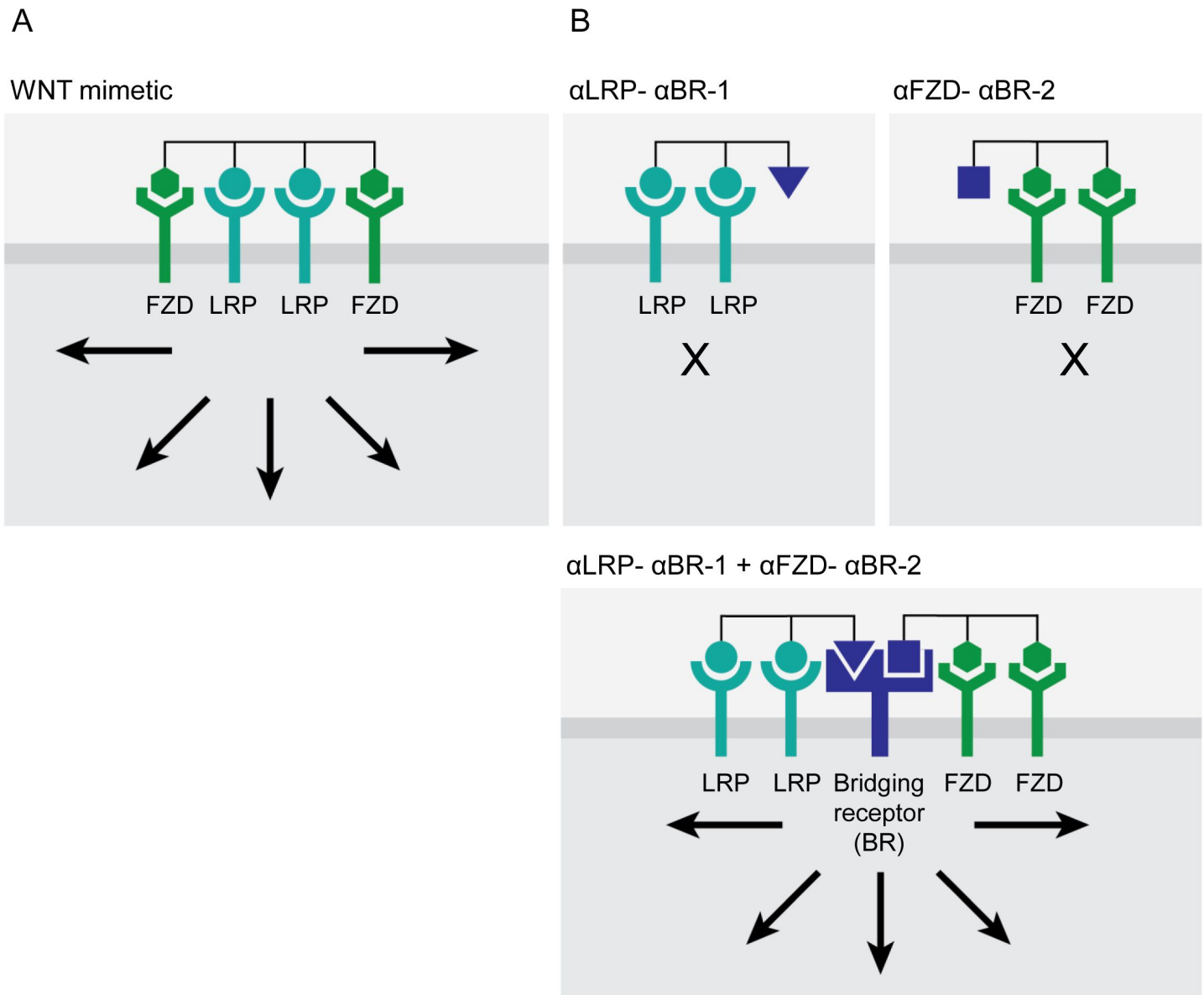


Figure 1.

Diagram of the activation of the signaling pathway by the split molecules through the binding to a bridging receptor on the targeting cell surface.

A) Efficient WNT/ β -catenin signaling requires two FZD binding domains and two LRP binding domains in one WNT mimetic molecule. B) WNT mimetic molecule is split into one molecule having two FZD binding arms (2:0) and a second molecule having two LRP binding arms (0:2) and then each tethered to a different “bridging element” binding to a different epitope on the same bridging receptor (named α LRP- α BR-1 and α FZD- α BR-2) (top two panels). On targeting cells where the BR is expressed, FZD and LRP are then assembled by α LRP- α BR-1 and α FZD- α BR-2 via BR recreating the signaling competent receptor complexes (bottom panel).

FGF21FL (full length FGF21) that can bind to β Klotho and competent to induce FGFR signaling;

FGF21 Δ C (FGF21 without the C-terminal β Klotho interaction domain) that does not bind β Klotho nor capable of inducing FGFR signaling;

FGF21 Δ N (FGF21 without the N-terminal FGFR interaction domain) that binds β Klotho but does not signal;

FGF21 Δ N Δ C that cannot bind β Klotho and cannot signal; 39F7 IgG that binds β Klotho and can induce FGFR signaling.

The FZD and LRP binding domains selected are F (binds FZD_{1,2,5,7,8}) and L (binds LRP6) previously named as F1 and L2, respectively (Chen et al., 2020). The graphic representations of the binders and the various combinations are shown in **Figs. 2A-C**.

To test the concept in **Fig. 1B**, the FZD binder (F) was combined with two versions of bridging receptor (β Klotho) binder, F-FGF21FL and F-FGF21 Δ N (**Fig. 2B**), and the LRP binder (L) was combined with the other bridging receptor (β Klotho) binder 39F7 as L-39F7 (**Fig. 2B**).

We first confirmed the binding of the various fusion molecules shown in **Fig. 2B** and **2C** to their respective target proteins. All F and L containing molecules bound to either FZD₇ or LRP6 E3E4 fragments as expected (**Figs. 2D-J**). FGF21FL, FGF21 Δ N, and 39F7 bound to β Klotho, while FGF21 without its C-terminal domain (FGF21 Δ C and FGF21 Δ N Δ C) lost bindings to β Klotho (**Figs. 2D-J**). Therefore, the molecule formats of the various fusion proteins had no significant impact on the individual binding element's binding to their target receptors. Next, we performed the Octet binding assays to assess whether the F-FGF21 or L-39F7 allow simultaneous bindings to their target receptors. Sequential additions of the various F-FGF21 fusion molecules to the sensor surface, followed by FZD₇ CRD, and then β Klotho show that a stepwise increase in binding signal was observed with FZD₇ CRD, but only F-FGF21FL and F-FGF21 Δ N showed additional binding to β Klotho but not from F-FGF21 Δ C or F-FGF21 Δ N Δ C (**Fig. 2K**). This suggests that both F-FGF21FL or F-FGF21 Δ N are capable of simultaneous bindings to both FZD₇ and β Klotho receptors. Sequential additions of the various L, 39F7, and α GFP combinations to the sensor surface, followed by LRP6E3E4, and then β Klotho show that a stepwise increase in binding signal with L-39F7 but not the other negative control molecules, α GFP-39F7 or L- α GFP (**Fig. 2L**). This suggests that L-39F7 can simultaneously bind to both LRP6 and β Klotho receptors.

The ability of this set of molecules to activate WNT/ β -catenin signaling was assessed in WNT-responsive Huh7 and HEK293 Super TOP-FLASH (STF) reporter cells (Zhang et al., 2020). As shown in **Fig. 3B**, the combination of F-FGF21FL or F-FGF21 Δ N with L-39F7 resulted in WNT/ β -catenin signaling in the liver cell line, Huh7 cells which expresses the bridging receptor β Klotho (*KLB*), but not in 293 cells where β Klotho is not expressed (**Fig. 3A** and **3G**). This signaling depends on the presence of both FZD and LRP binding arms and the ability to bind the bridging receptor, as the removal of LRP binding arm L from L-39F7 or inactivation of β Klotho binding arms (use of FGF21 Δ C or FGF21 Δ N Δ C, or the replacement of 39F7 with α GFP) resulted in no activity in either cells (**Figs. 3A-F**). This provided experimental evidence supporting the SWIFT concept presented in **Fig. 1B**.

Validation of liver specific targeting with primary human cells

To test the potency of these molecules in cells directly derived from the tissue of interest we selected primary human hepatocytes and human small intestinal organoids (**Fig. 4A**). These cultures better represent the physiological and transcriptional characteristics of the liver and intestine compared to cell lines. As expected, primary human hepatocytes express bridging receptor β Klotho, while human small intestinal cells do not (**Fig. 4B**). The combination of F-

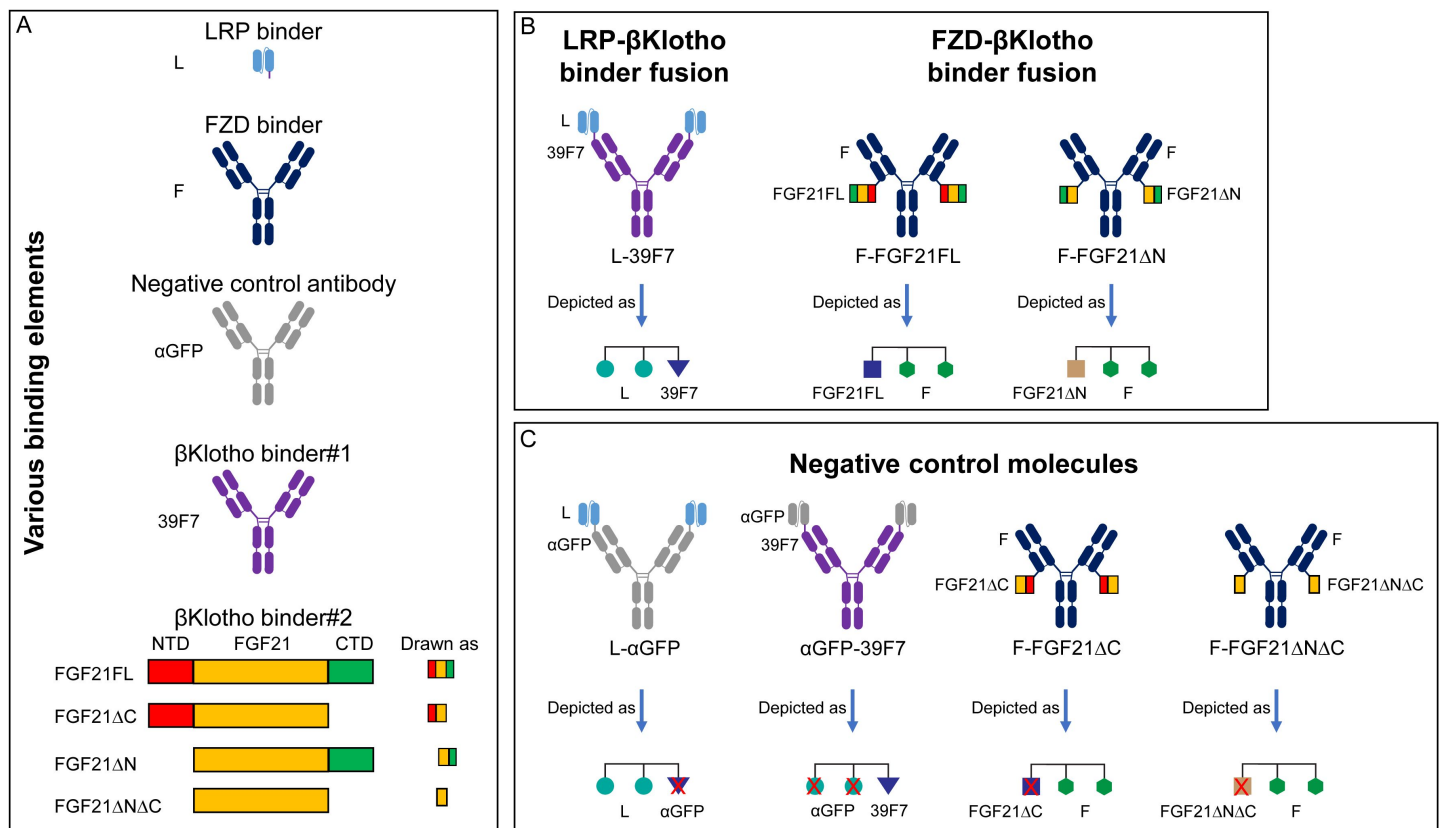


Figure 2.

Diagrams of the molecules used in the experiment to provide proof of concept for SWIFT.

A) Elements that are used in the split molecules. LRP6 binder element is in scFv format; FZD binder is in the IgG1 format; Both α GFP IgG1 and α GFP scFv are used for assembly of the negative control molecules. Two types of β Klotho binders are used. Binder#1, 39F7 IgG1, a β Klotho monoclonal antibody; Binder#2, FGF21FL and different deletion variants. B) The diagram of the assembled SWIFT molecules. C) The diagram of the assembled negative control molecules. (D-G) Bindings of various F-FGF21 fusion proteins to FZD₇ and β Klotho. Bindings of FZD₇ and β Klotho to F-FGF21FL (D), F-FGF21 Δ N (E), F-FGF21 Δ C (F), or F-FGF21 Δ N Δ C (G) were determined by Octet. (H-J) Binding of LRP6 and β Klotho to L-39F7 (H), α GFP-39F7 (I), or L- α GFP (J) were determined by Octet. Mean K_D values were calculated from all 7 binding curves with global fits (red dotted lines) using 1:1 Langmuir binding model. (K) Step bindings of FZD₇ and β Klotho to various F-FGF21 fusion proteins. Sequential binding of F-FGF21FL (blue sensorgram), F-FGF21 Δ N (red sensorgram), F-FGF21 Δ C (light green sensorgram), or F-FGF21 Δ N Δ C (green sensorgram), followed by FZD₇ CRD, then followed by addition of β Klotho on Octet shows simultaneous engagement of both FZD₇ and β Klotho to the indicated F-FGF21 proteins. Sensorgrams for FZD₇ and β Klotho area (red dotted box) are enlarged at the right. (L) Step binding of LRP6 and β Klotho to L-39F7 and its control proteins. Sequential binding L-39F7 (blue sensorgram), L- α GFP (light blue sensorgram), α GFP-39F7 (red sensorgram), or 39F7 IgG (green sensorgram), followed by LRP6E3E4, then followed by addition of β Klotho on Octet shows simultaneous engagement of both LRP6 and β Klotho to the indicated L-39F7 and its control proteins. Sensorgrams for LRP6 and β Klotho area (red dotted box) are enlarged on the right.

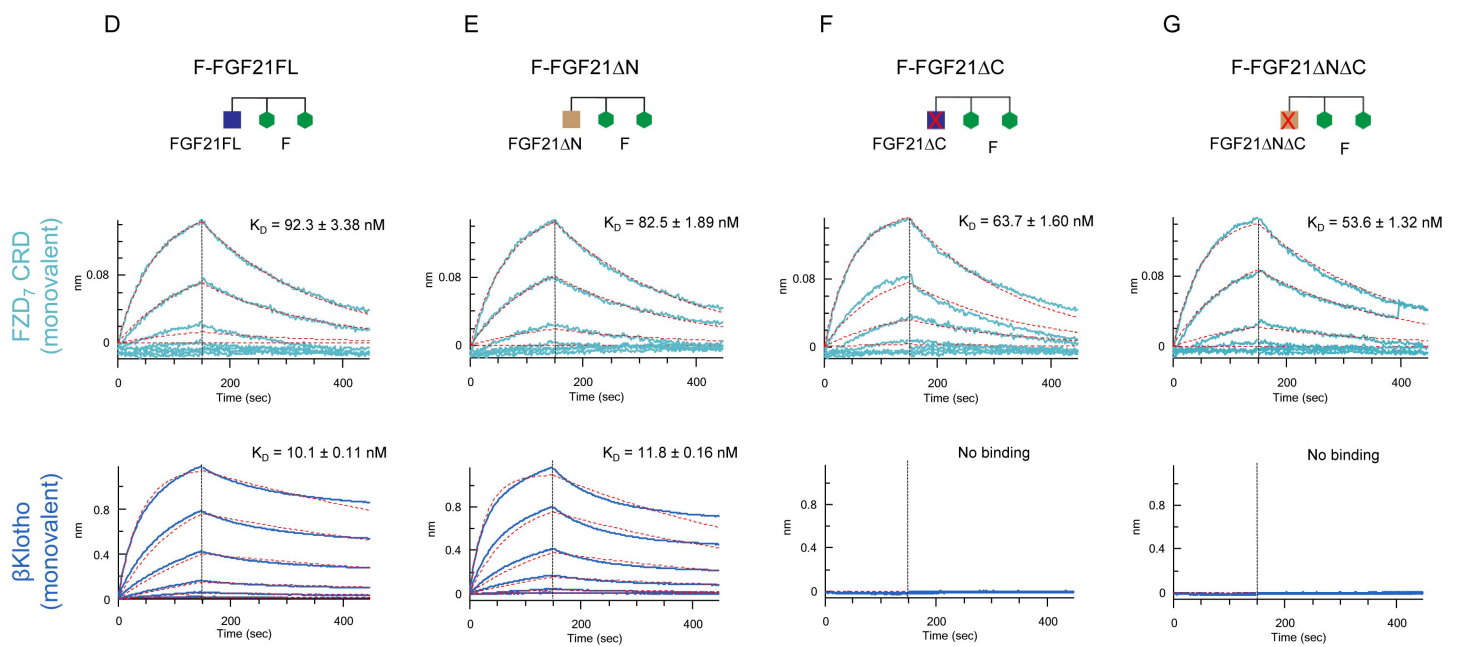


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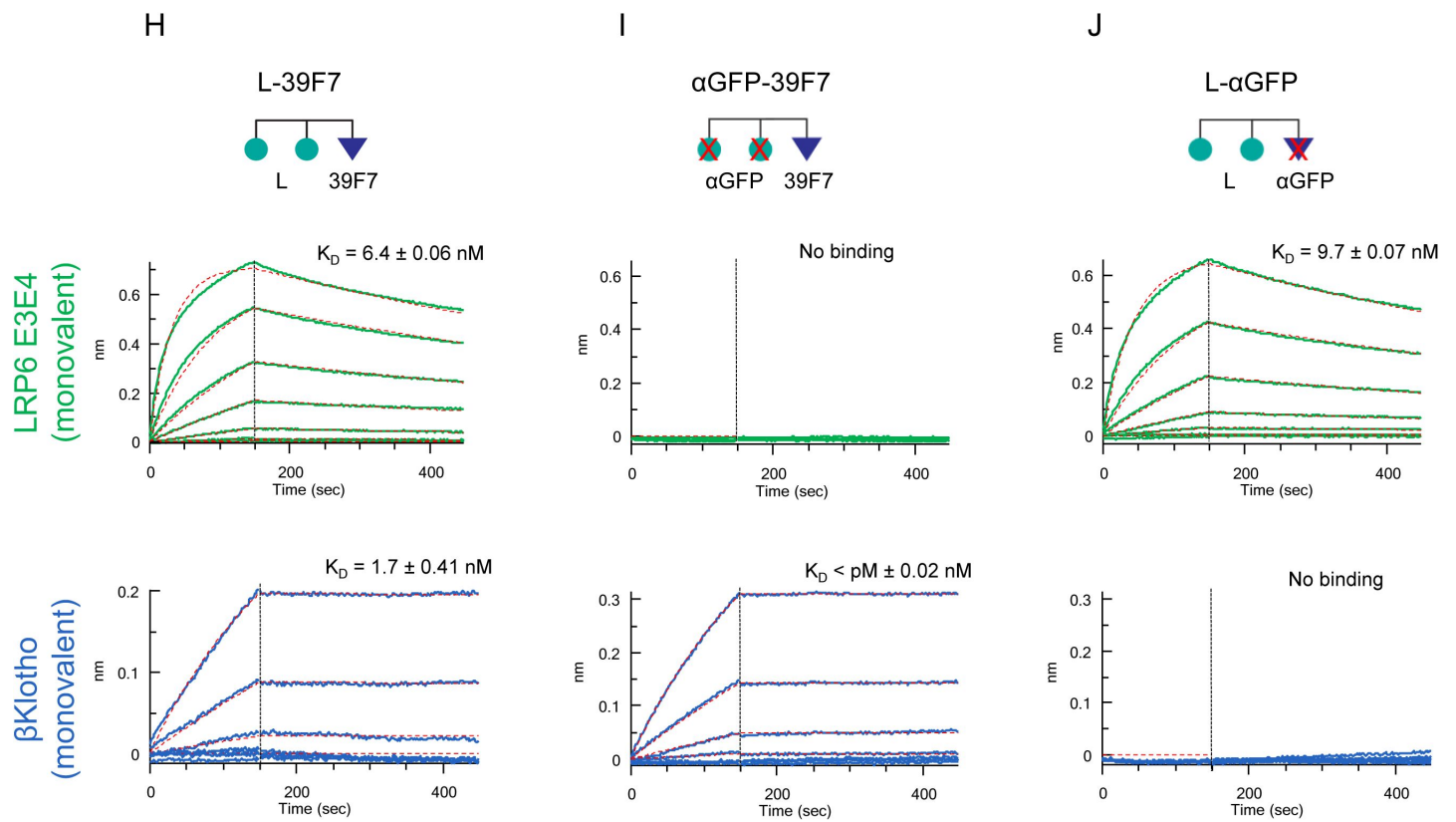
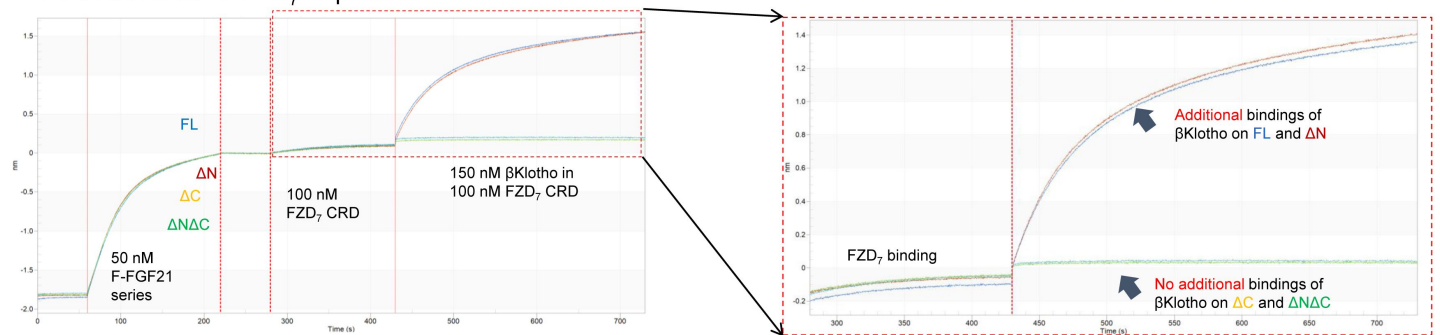


Figure 2. (continued)

K

F-FGF21 series $\rightarrow$ FZD₇ $\rightarrow$ β Klotho



L

L-39F7 series $\rightarrow$ LRP6 $\rightarrow$ β Klotho

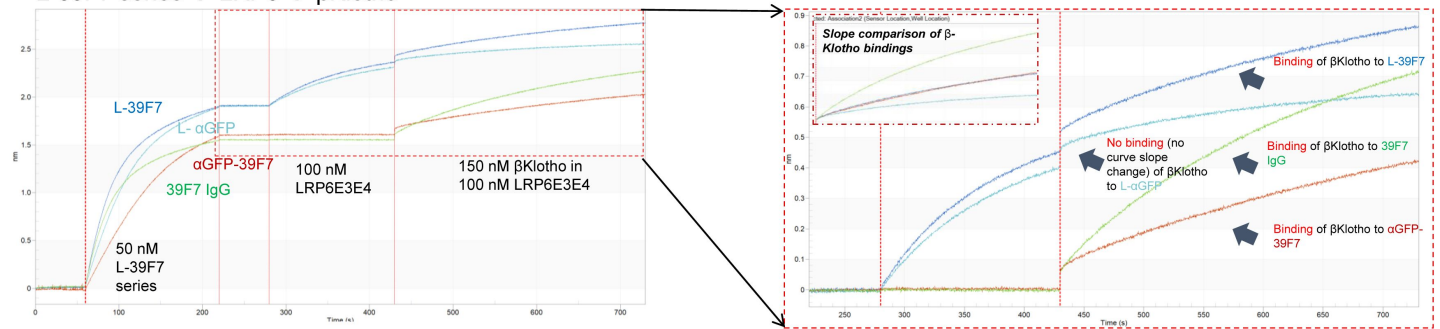


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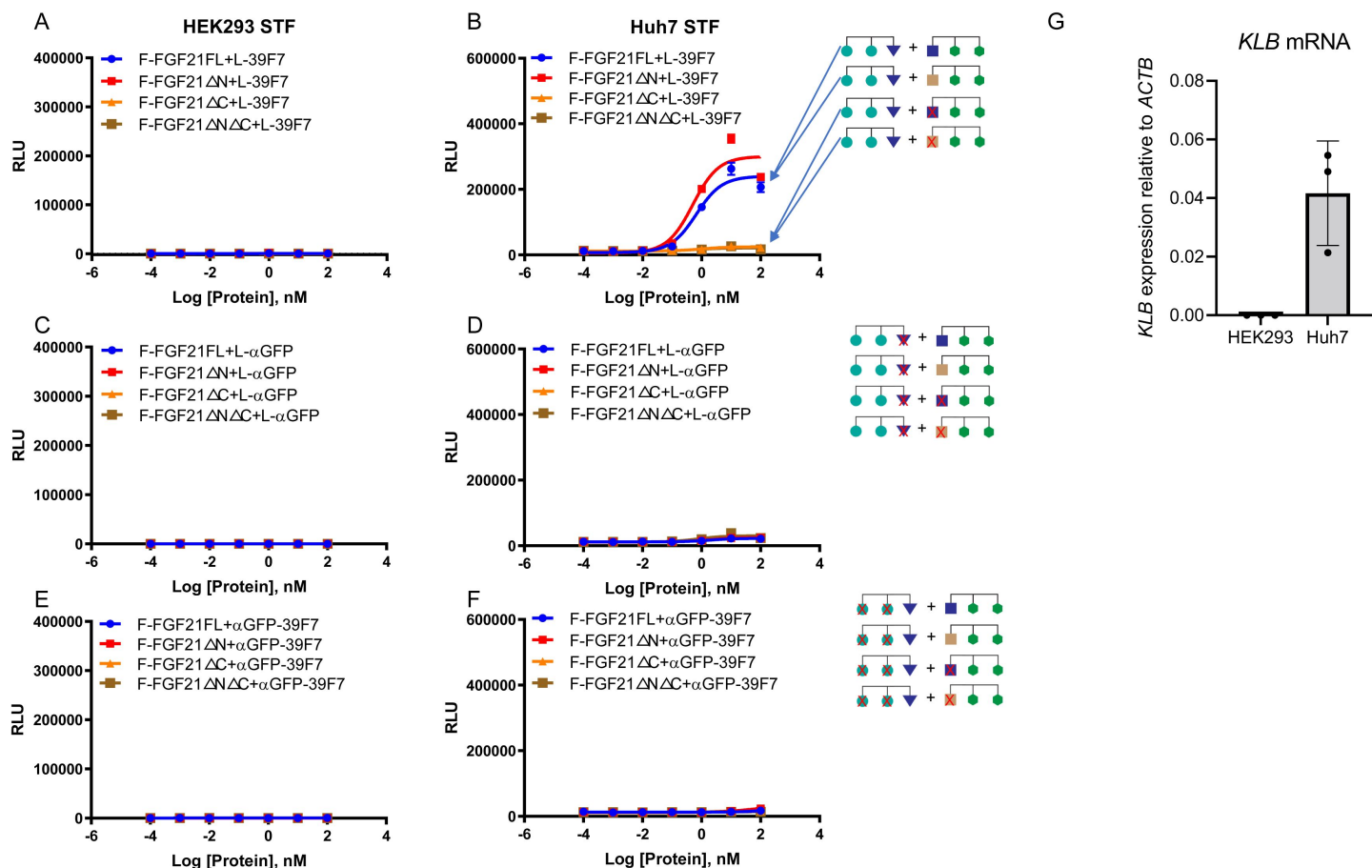


Figure 3.

Dose dependent STF assay of SWIFT molecules in HEK293 and Huh7 cells.

A-F) Various F-FGF21 fusion proteins in the presence of L-39F7 in HEK293 STF cells (A) and Huh7 STF cells (B); various F-FGF21 fusion proteins in the presence of L-αGFP in HEK293 STF cells (C) and Huh7 STF cells (D); and various F-FGF21 fusion proteins in the presence of αGFP-39F7 in HEK293 STF cells (E) and Huh7 STF cells (F). G) Expression of bridging receptor βKlotho (*KLB*) in HEK293 and Huh7 cells. Data are representative of three independent experiments performed in triplicates and are shown as mean ± standard deviation (SD).

FGF21ΔN and L-39F7 in human hepatocytes had a robust increase in the expression of WNT target gene *AXIN2* after 24 hours compared to the control (**Fig. 4C**). The combination with negative control F-FGF21ΔNΔC did not result in the elevation of *AXIN2*. The same panel of molecules did not elicit any target gene expression in human small intestinal cells, confirming the need for bridging receptor expression for activity (**Fig. 4D**). These results in primary cells support the feasibility of cell specific targeting, and we expect them to be predictive of tissue specific *in vivo* responses.

Discussion

A major deficiency in therapeutic development today is effective approaches to target drugs toward a desired cell type. Due to the pleiotropic expression and actions of most drug targets in many cell types/organs or lack of differentiation between diseased vs normal cells, many drugs exhibit either dose limiting toxicity or act on multiple cell types of opposing activity that render drugs less effective. Therefore, developing effective cell targeting methods would allow reduced systemic toxicity and higher efficacy for more effective treatments.

Here we describe a novel cell targeting approach we termed BRAID where an active drug molecule is divided into inactive parts that are assembled via a bridging receptor specific to the target cell. Although complementation and splitting a molecule into inactive parts have been explored previously, the novelty of our approach is that there is no requirement for the two split inactive components to interact with one another and therefore, does not require cooperative interactions between the split components and receptors for activity. The assembly of the signaling complex by our approach is achieved by simultaneous binding of the two divided inactive components through a common bridging receptor (**Fig. 1B**).

We tested this concept on WNT/β-catenin signaling pathway as a model system. This signaling system requires two receptors, FZD and LRP. We, and others, have generated antibody based WNT mimetic molecules by linking FZD binders and LRP binders into a single molecule to activate the FZD/LRP receptor complex (Chen et al., 2020; Janda et al., 2017, Tao et al., 2019). To explore the BRAID concept with the WNT signaling receptors, we first divided the tetravalent bispecific WNT mimetic into two components: one contains two FZD binding domains and the other contains two LRP binding domains. To each of these two inactive components, a binding moiety to the bridging receptor, βKlotho (predominantly expressed in hepatocytes), was attached. The two bridging receptor binding moieties are FGF21 and 39F7 which bind to non-overlapping regions on βKlotho (Min et al., 2018). As demonstrated both on WNT responsive reporter cell lines as well as on primary human hepatocytes and human intestinal organoids, target cell specific WNT signaling was indeed observed when the two inactive components were combined. The signaling only occurred on the target cell hepatocytes, but not on HEK293 nor intestinal cells. Furthermore, activity depends on the presence of βKlotho binding domains on the two inactive components, as removal of βKlotho binding on either the FZD or the LRP half of the molecule or having βKlotho but not LRP binding, resulted in inactive combinations (**Figs. 3** and **4**). These results provided compelling evidence for this novel cell targeting concept. We envision that other potential ways of dividing the WNT or the WNT mimetic molecules, including different geometry, or stoichiometry of FZD/LRP ratios, linker length, or having one each of FZD and LRP binding arms as one component may also result in active targeted WNT mimetics. We have termed this cell targeted WNT system, SWIFT. As WNT plays important roles in many tissues during development and in adult tissue homeostasis and injury repair, and dysregulated WNT signaling may result in various human diseases (Nusse and Clevers, 2017), SWIFT offers the opportunity to more specifically target WNT activation in desired cell types that can help both research and therapeutic development.

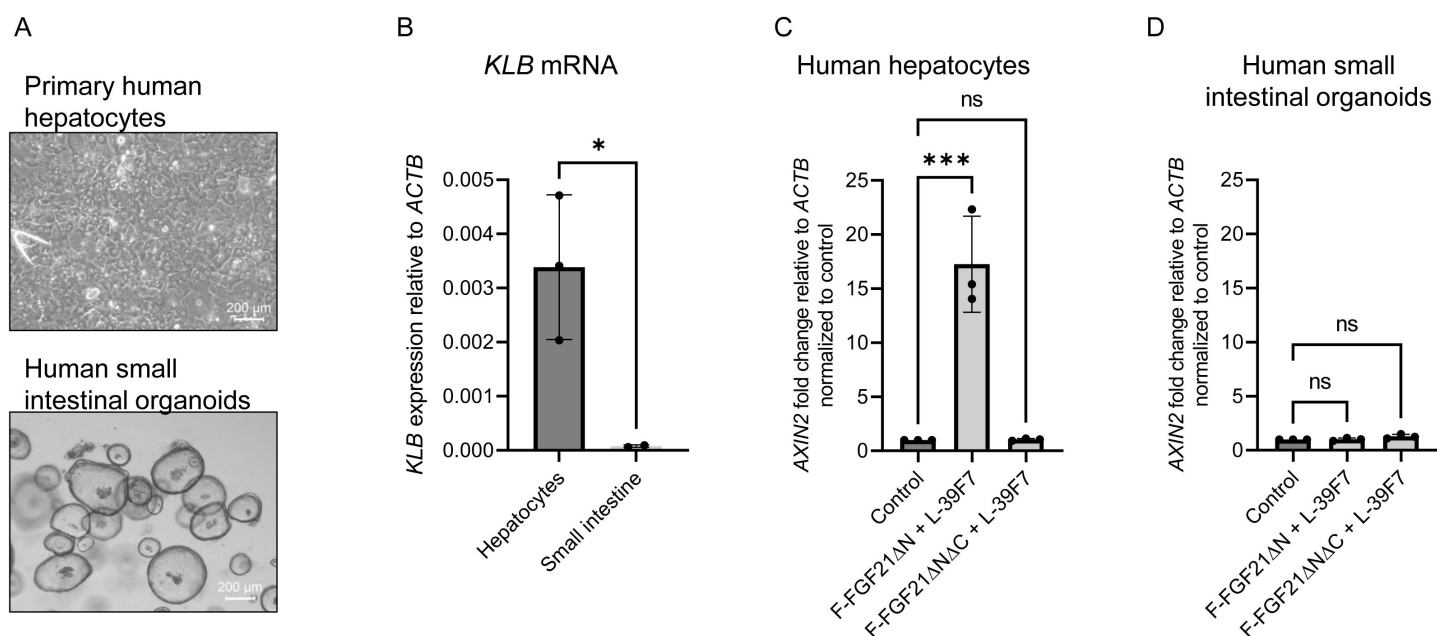


Figure 4.

Activity of targeted molecules in primary human cells.

A) Representative images of primary human hepatocytes cultures in 2D or human small intestinal organoids. Scale bars 200 μ m. B) Expression of bridging receptor β Klotho (*KLB*) in hepatocytes or small intestinal cells. C) WNT target gene *AXIN2* expression normalized to control treatment after 24-hour treatment with 10 nM of indicated molecules in human hepatocytes. D) WNT target gene *AXIN2* expression normalized to control treatment after 24-hour treatment with 10 nM of indicated molecules in human small intestinal organoids. *** = $P \leq 0.001$ (one-way ANOVA), each data point represents an independent experiment performed in duplicates.

In conclusion, we describe here a novel cell targeting approach that can be broadly applied to various signaling systems. This approach also allows the combination of different signaling pathways together resulting in additive or synergistic effects.

Acknowledgements

The authors would like to thank Leona Cheng, Haili Zhang, Jasmine Tan, Hayoung Go, and Sean Bell for technical support and discussions. We thank Huy Nguyen for the illustration in **Figure 1** [↗](#). And we thank Wen-Chen Yeh and Craig Parker for critical reading of the manuscript and Kathee Littrell for editorial support.

Author contributions

HC, SJL and YP Formal analysis, Supervision, Investigation, Visualization, Methodology, Writing—review and editing; HC, SJL, RL, AS, NS, AD, YP, MV, CL and YP, Resources, Investigation, Visualization, Methodology, Writing—review and editing; YL, Conceptualization, Formal analysis, Resources, Supervision, Funding acquisition, Visualization, Methodology, Writing—original draft; Writing—review and editing

Declaration of interests

All authors are current or former full-time employees and shareholders of Surrozen, Inc. YL is Senior Vice President, Biology at Surrozen, Inc. A patent application is pending for the work described in this manuscript.

Materials and methods

Molecular Cloning

All constructs were cloned into pcDNA3.1(+) mammalian expression vector (Thermo Fisher). The L-scFv binder was constructed by fusing the heavy-chain variable region (VH) to the N-terminus of the light-chain variable (VL) region with a 15-mer linker (GSAASGSSGGSSGA); The anti-GFP scFv binder was constructed by fusing the VH to the N-terminus of the VL with a 15-mer linker (GGGSGGGSGGGGS). All human IgG1 constructs contain the L234A/L235A/P329G mutations (LALAPG) in Fc domain to eliminate effector function (Lo et al., 2017 [↗](#)). For generating the constructs of scFv-39F7 IgG1, the scFv binder was fused to the N-terminus of 39F7 LC with a 5-mer-linker GSGSG. For generating the construct of L-αGFP IgG1, the L-scFv was fused to the N-terminus of αGFP LC with a 15-mer-linker GSGSGSGSGSSGG. For generating the FGF21 variants appended F IgG molecules, FGF21 variants were fused to the C-terminus of the LC of F IgG1 (LALAPG) with a 5-mer-linker (GGSGS). FGF21FL has the mature protein sequence (H29-S209) with RGE mutations in the C-terminus (Stanislaus et al., 2017 [↗](#)); FGF21ΔN is the sequence of FGF21FL with the deletion of the N-terminus of sequence of H29-R45; FGF21ΔC is the sequence of FGF21FL with the deletion of the C-terminus S190-S209; FGF21ΔNΔC is the sequence of FGF21FL with the deletion of both the N-terminal sequence of H29-R45 and the C-terminal sequence of S190-S209.

Protein Production

All recombinant proteins were produced in Expi293F cells (Thermo Fisher Scientific) by transient transfection. The proteins were first purified using CaptivA Protein A affinity resin (Repligen), unless otherwise specified. All proteins were further polished with Superdex 200 Increase 10/300 GL (GE Healthcare Life Sciences) size-exclusion chromatography (SEC) using 1 x HBS buffer (20 mM HEPES pH 7.4, 150 mM NaCl). The proteins were subsequently examined by SDS-polyacrylamide electrophoresis and estimated to have > 90% purity.

Super Top-Flash (STF) Assay

WNT signaling activity was measured using HEK293 and Huh7 cells containing a luciferase gene controlled by a WNT-responsive promoter (STF assay) as previously reported [Zhang et al., 2020](#). In brief, cells were seeded at a density of 10,000 per well in 96-well plates 24 hours prior to treatment at the presence of 3 μ M IWP2 to inhibit the production of endogenous WNTs and the presence of 20 nM Fc-Rspodin 2. Cells were lysed with Luciferase Cell Culture Lysis Reagent (Promega) and luciferase activity was measured with Luciferase Assay System (Promega) using vendor suggested procedures.

Affinity Measurement and Step-Binding Assay

Binding kinetics of F-FGF21 series (F-FGF21FL, F-FGF21 Δ N, F-FGF21 Δ C, and F-FGF21 Δ N Δ C) to human FZD₇ CRD and β Klotho (Fisher Scientific) or L-39F7 series (L-39F7, α GFP-39F7, and L- α GFP) to human LRP6E3E4 and β Klotho, respectively, were determined by bio-layer interferometry (BLI) using an Octet Red 96 (PALL ForteBio) instrument at 30°C, 1000 rpm with AHC biosensors (Sartorius). Various F-FGF21 or L-39F7 proteins were diluted to 50 nM in the running buffer and captured to the AHC biosensor, followed by dipping into wells containing the FZD₇ CRD, LRP6E3E4 and β Klotho at different concentrations in a running buffer or into a well with only the running buffer as reference channel. The dissociation of the interaction was followed with the running buffer. The monovalent K_D for each binder was calculated by Octet System software, based on fitting to a 1:1 binding model.

Step-binding assay was performed with the BLI using the Octet Red 96 instrument at 30°C, 1000 rpm with AHC biosensors. Various F-FGF21 or L-39F7 proteins were diluted to 50 nM in the running buffer and captured to the AHC biosensor, followed by dipping into wells containing the 100 nM FZD₇ CRD or 100 nM LRP6E3E4, respectively. The sensor chips next moved into 150 nM β Kloths containing 100 nM FZD₇ CRD or containing 100 nM LRP6E3E4 to check the additional bindings of β -Klotho. Sensorgram slopes were compared for β Klotho bindings.

Primary human cells

Human hepatocytes were purchased from BioIVT (10-donor pooled cryoplateable X008001-P) and cultured in LONZA hepatocyte maintenance medium (CC-3198). In short, plastic culture plates were coated with 20% Matrigel Matrix (CB40230C) and cells were plated in plating medium (BioIVT Z990003). After four hours the medium was changed to maintenance medium and refreshed every day for three days prior to the 24h experiment.

Human small intestinal organoids were a gift from the Calvin Kuo Lab at Stanford. Organoids were maintained and expanded as previously described ([Sato et al., 2011](#)). In short, adapted expansion medium contained Advanced DMEM, 10 mM HEPES, 1x GlutaMAX, 1X Penicillin-Streptomycin, 1x B27, 1x N2, 1.25 mM N-acetylcysteine, 10 mM Nicotinamide, 50 ng/mL recombinant human EGF, 50 ng/mL recombinant human Noggin, 20 nM R-Spondin 2, 0.1 nM L-F Wnt mimetic, 10 nM recombinant Gastrin, 500 nM A83-01 and 10 μ M SB202190.

Treatment of molecules was done in the presence of 20 nM R-spondin 2 for 24h at a concentration of 10 nM. After 24 hours the cells were harvested and RNA collected for qPCR. Each experiment with both primary human hepatocytes and human small intestinal organoids was repeated three times.

Quantitative polymerase chain reaction analysis of gene expression

RNAs from HEK293, Huh7 cells, or primary human cells were extracted using the Qiagen RNeasy Micro Kit (Qiagen). cDNA was produced using the SuperScript IV VILO cDNA Synthesis Kit (Thermo Fisher). β Klotho (*KLB*) RNA was quantified using Maxima SYBR Green qPCR master mix on a Bio-Rad CFX96 real time PCR machine.

Cycle threshold (C_t) values were normalized to the expression of constitutive *ACT1NB* RNA using the following oligo's: ACTB_F1: CTGGAACGGTGAAGGTGACA. ACTB_R1: AAGGGACTTCCTGTAACAATGCA. KLB_F1: ATCTAGTGGCTTGGCATGGG. KLB_R1: CCAAACCTTCGAGTGAGCCTTG. KLB_F2: CACTGAATCTGTTCTTAAGCCCG. KLB_R2: GGCGTTCCACACGTACAGA. KLB_F3: GGAGGTGCTGAAAGCATACCT. KLB_R3: TCTCTTCAGCCAGTTTGAATGC

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Reviewer #1 (Public Review):**Summary:**

The goal of this study was to develop and validate novel molecules to selectively activate a cell signaling pathway, the Wnt pathway in this case, in target cells expressing a specific receptor. This was achieved through a two-component system that the authors call BRAID, where each component simultaneously binds the target cell-specific marker BKlotho and a Wnt co-receptor. These components, called SWIFT molecules, bring together the Wnt co-receptors LRP and FZD, activating the pathway specifically in cells that express BKlotho. Results presented in the study demonstrate the desired activity of SWIFT molecules; the binding assays support the simultaneous association of SWIFT with BKlotho and a Wnt co-receptor, and the Wnt reporter and qPCR assays support pathway activation in cell lines and primary cells in a BKlotho-dependent manner. In the future, the BRAID approach could be applied to activate Wnt signaling or another pathway initiated by a co-receptor complex in a cell type-specific manner, and/or in a FZD subtype-specific manner to activate distinct branches of Wnt signaling.

Strengths:

- This study successfully demonstrates a novel way to activate Wnt signaling in target cells expressing a specific marker. Given the role of the Wnt signaling pathway in key processes such as cell proliferation and tissue renewal and the value of modulating cell signaling in a cell type-specific manner, the cell targeting system developed here holds great therapeutic and research potential. It will be curious to see whether the BRAID design can be applied to other cell surface markers for Wnt activation, or for activation of other signaling pathways that require co-receptor association.
- Octet assay results show simultaneous binding of SWIFT molecules to both the Wnt co-receptor FZD/LRP and BKlotho, while negative control molecules without the FZD/LRP or BKlotho-binding module show neither receptor binding nor Wnt pathway activation. These results indicate that SWIFT molecules function through the intended mechanism.

Weaknesses:

- Here, the activity of SWIFT molecules was assessed in single cell types with or without BKlotho expression. Ultimately, the ability of the SWIFT molecules to activate Wnt signaling

in a cell type-specific manner should be tested in the context of many different cellular identities that express BKlotho to different extents. It would be good to demonstrate that Wnt activation by SWIFT correlates with BKlotho expression level in multiple cell types - such data would strengthen the claim of cell-type specificity.

- The study does not address whether the targeted cells express FGFR1c/2c/3c and whether the FGF21 full-length moiety or the 39F7 IgG moiety of SWIFT molecules could unintentionally activate FGF signaling in these cells.

Reviewer #2 (Public Review):

Summary:

The study introduces BRAID, a novel approach for targeting drugs to specific cell types, addressing the challenges of pleiotropic drug actions. Unlike existing methods, this one involves breaking a protein drug molecule into inactive parts that are then put back together using a bridging receptor on the target cell. The individual components of this assembly are not required to be together, thereby affording it a degree of flexibility. The authors applied this idea to the WNT/-catenin signaling pathway by splitting a WNT mimic into two parts with FZD and LRP binding domains and bridging receptors. This combined method, which is called SWIFT, showed that WNT signaling was turned on in target cells, showing cell-specific targeting. The technique shows promise for the development of therapeutics, as it provides a way to more precisely target signaling pathways.

The authors have effectively elucidated their strategy through visually appealing diagrams, providing clear and thorough visual aids that facilitate comprehension of the concept. In addition, the authors have provided convincing evidence that the C-terminal region of FGF21 is essential for the binding process. Their meticulous and thorough presentation of experimental results emphasizes the significance of this specific binding domain and validates their findings.

Strengths:

BRAID, a novel cell targeting method, divides an active drug molecule into inactive components formed by a bridging receptor. This novel approach to cell-specific drug action may reduce systemic toxicity.

The SWIFT approach successfully targets cells in the WNT/ β -catenin signaling pathway. The approach activates WNT signaling only in target cells (hepatocytes), proving its specificity.

The study indicates that the BRAID approach can target various signaling systems beyond WNT/ β -catenin, indicating its versatility. Therapeutic development may benefit from this adaptability.

Weaknesses:

The study shows the SWIFT approach works in vitro using cell lines, primary human hepatocytes, and human intestinal organoids, but it lacks an in vivo animal model or clinical validation. The applicability of this approach to therapy is still unknown.

The success of SWIFT depends on the presence and expression of the bridging receptor (β Klotho) on target cells. The approach may fail if the target receptor is not expressed or available.